

THE CUTTING EDGE OF CONNECTIVITY: GaN-on-SiC Amplifiers For Satellites, Radars, And 5G



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Power, efficiency, and reliability are non-negotiable for satellites, radars, and 5G networks. GaN-on-SiC amplifiers deliver on all fronts.

The growing complexity of military satellites, radar technologies, and 5G networks calls for advanced solutions to tackle modern communication challenges. Gallium nitride (GaN) on silicon carbide (SiC) monolithic microwave integrated circuit (MMIC) power amplifiers stand out, offering exceptional power efficiency, thermal management, and operational bandwidth—making them indispensable for next-generation communication systems.

Key challenges in modern communication systems

High-power requirements. Military satellites, which often operate over vast distances, and radar systems, used for precise tracking and

detection, require high-output power. Similarly, 5G networks rely on robust amplifiers to maintain ultra-high-speed data transmission and connectivity. Traditional power amplifiers struggle to deliver such high power efficiently, leading to performance bottlenecks.

Thermal management issues. The generation of significant heat during high-power operation is a critical challenge. If not managed effectively, heat can impair device performance, shorten life span, and reduce reliability.

Broad frequency bandwidth needs. Modern military and commercial communication systems require amplifiers that can operate across wide frequency ranges without compromising signal clarity or

